

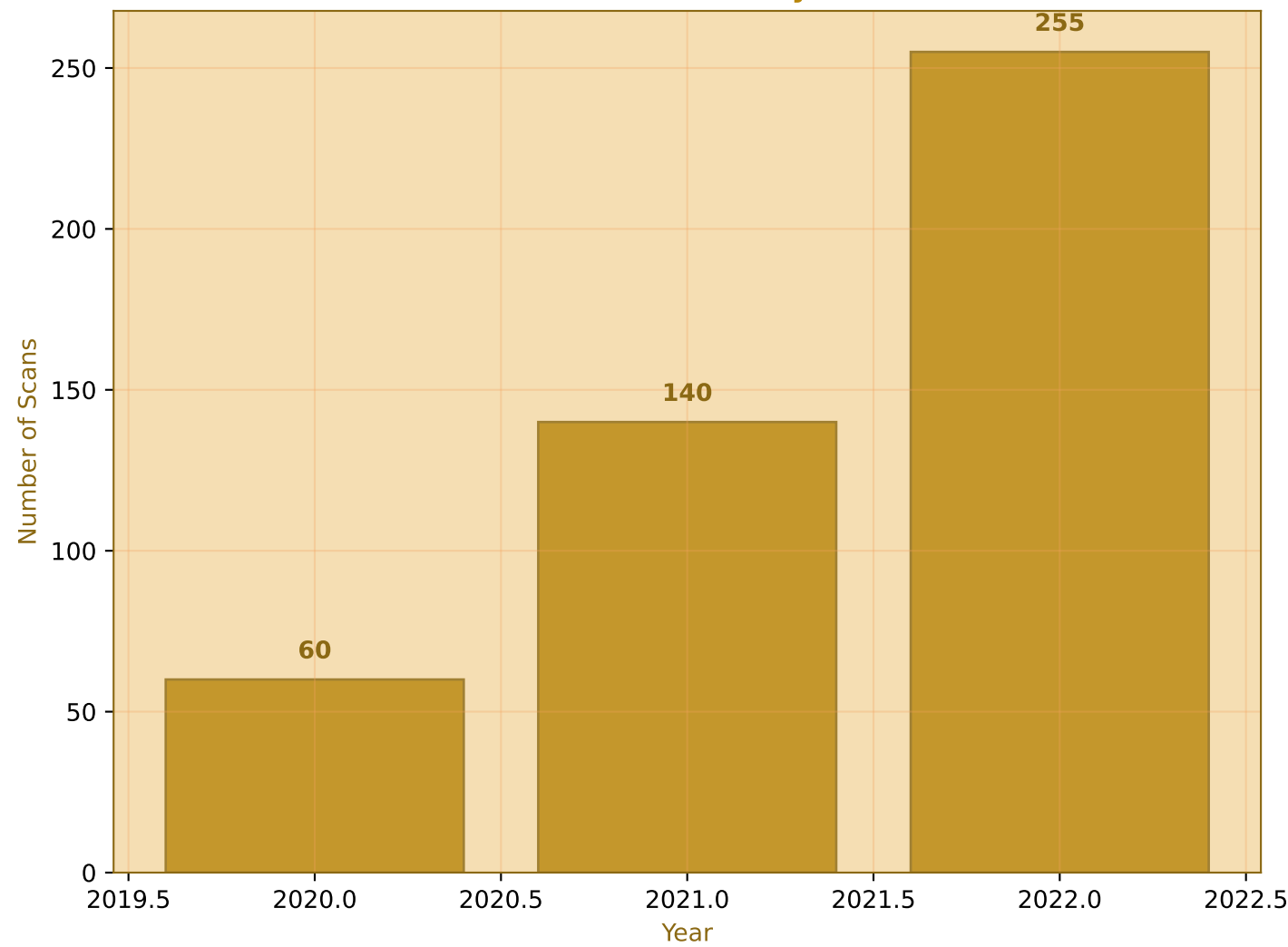
Machine Learning Using Real Radiomics Data Feature Analysis for Outcome Prediction 2020-2022 Dataset Analysis

Real Data Implementation with 80-20 Split

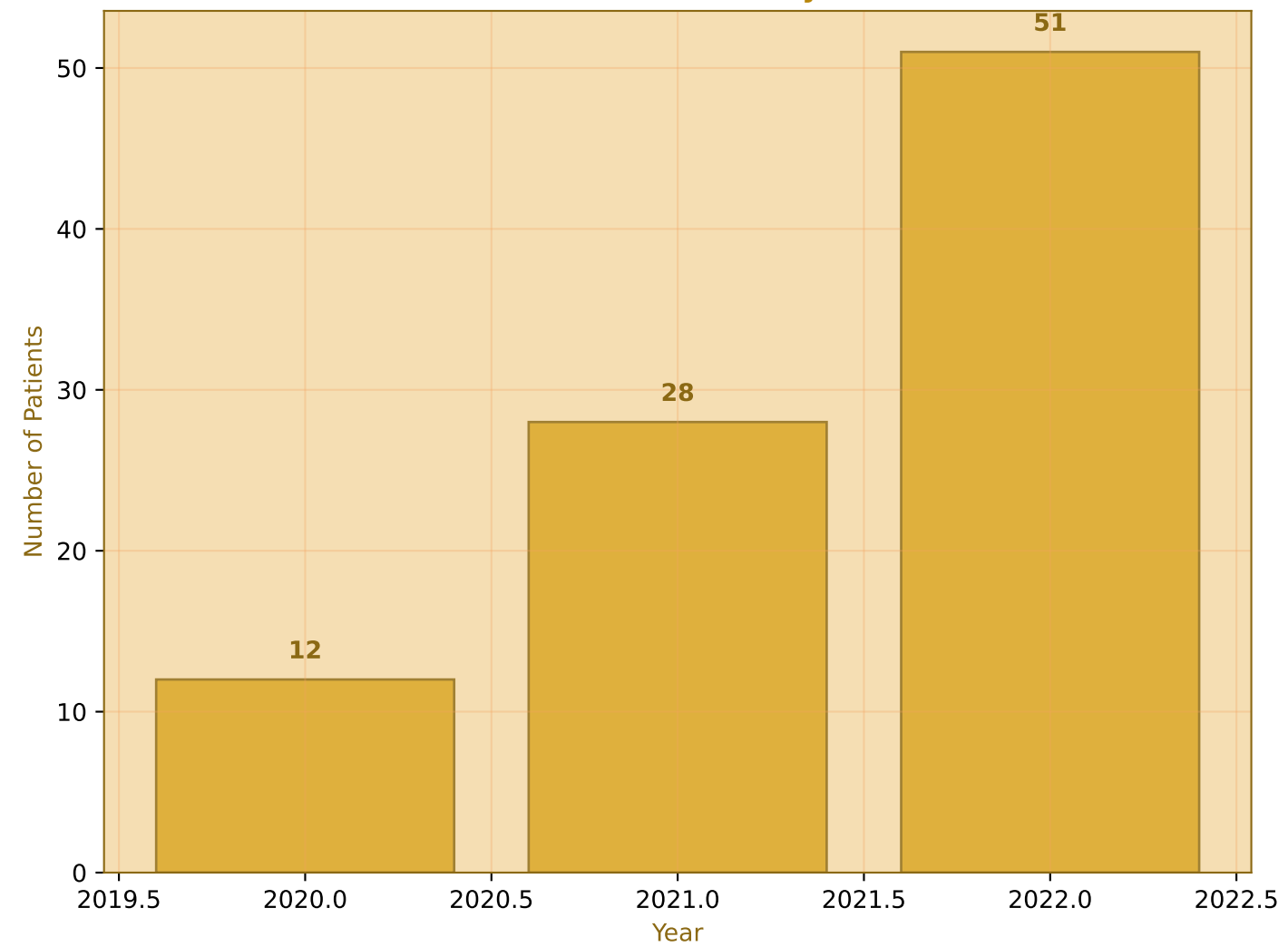
- **Real Dataset Overview (2020-2022)**
 - **Patient and Scan Distribution by Year**
 - **Outcome Distribution Analysis**
 - **Machine Learning Model Training (80-20 Split)**
 - **Model Performance Evaluation**
 - **Feature Importance Analysis**
 - **Year-wise Performance Comparison**
- Dataset Analysis: 70 patients, 455 scans, 127 features

Real Dataset Overview (2020-2022)

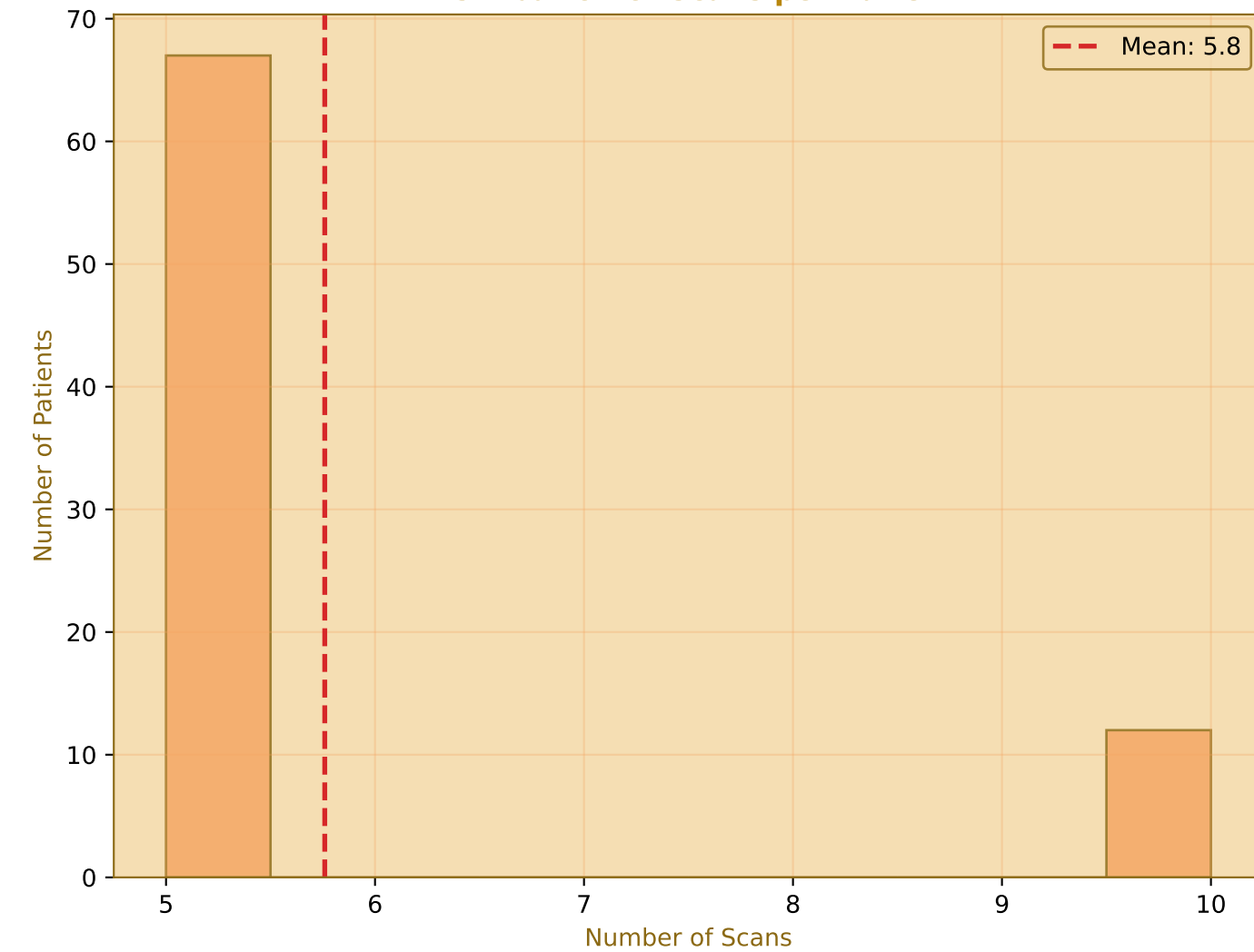
Number of Scans by Year



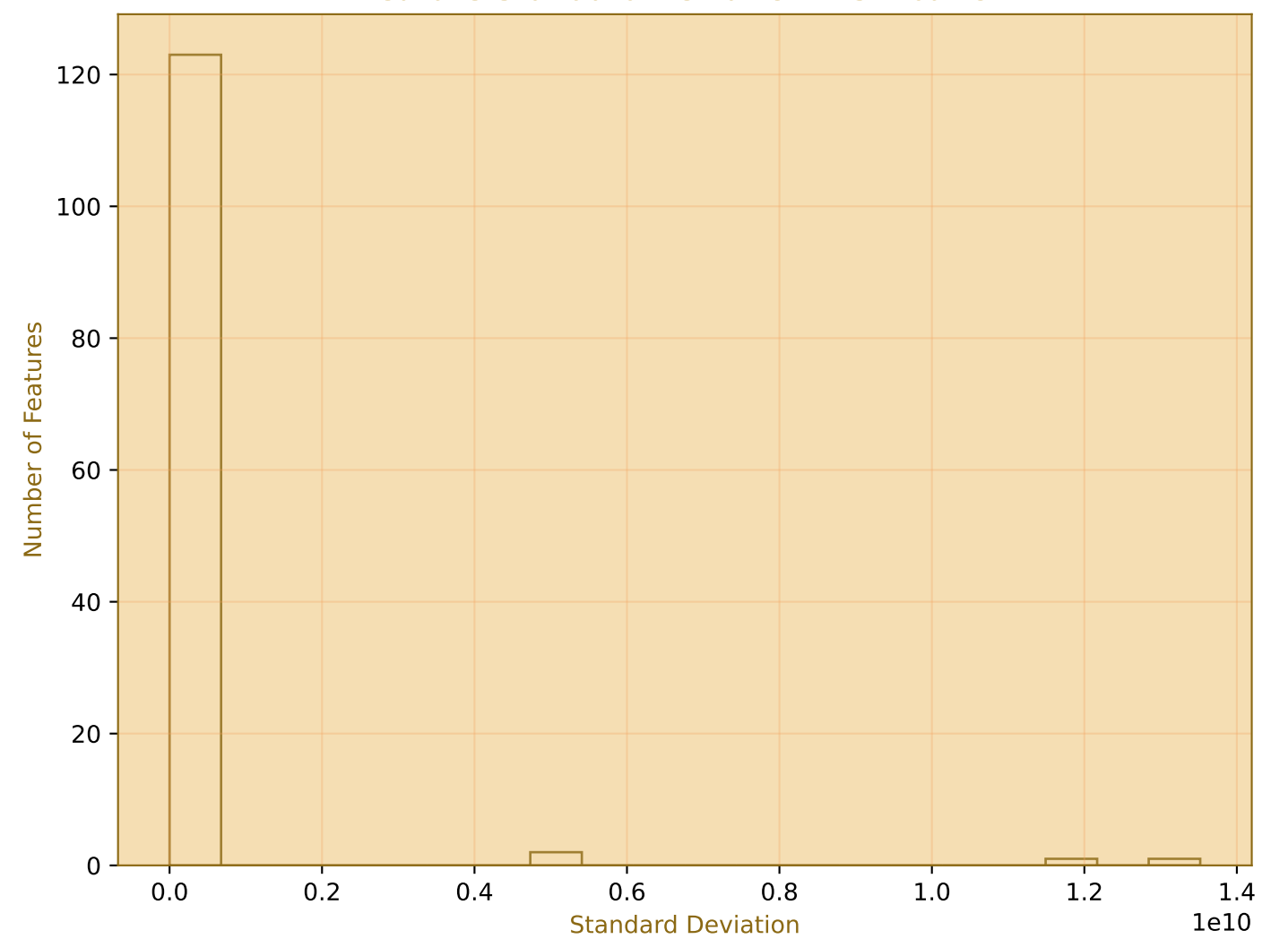
Number of Patients by Year



Distribution of Scans per Patient

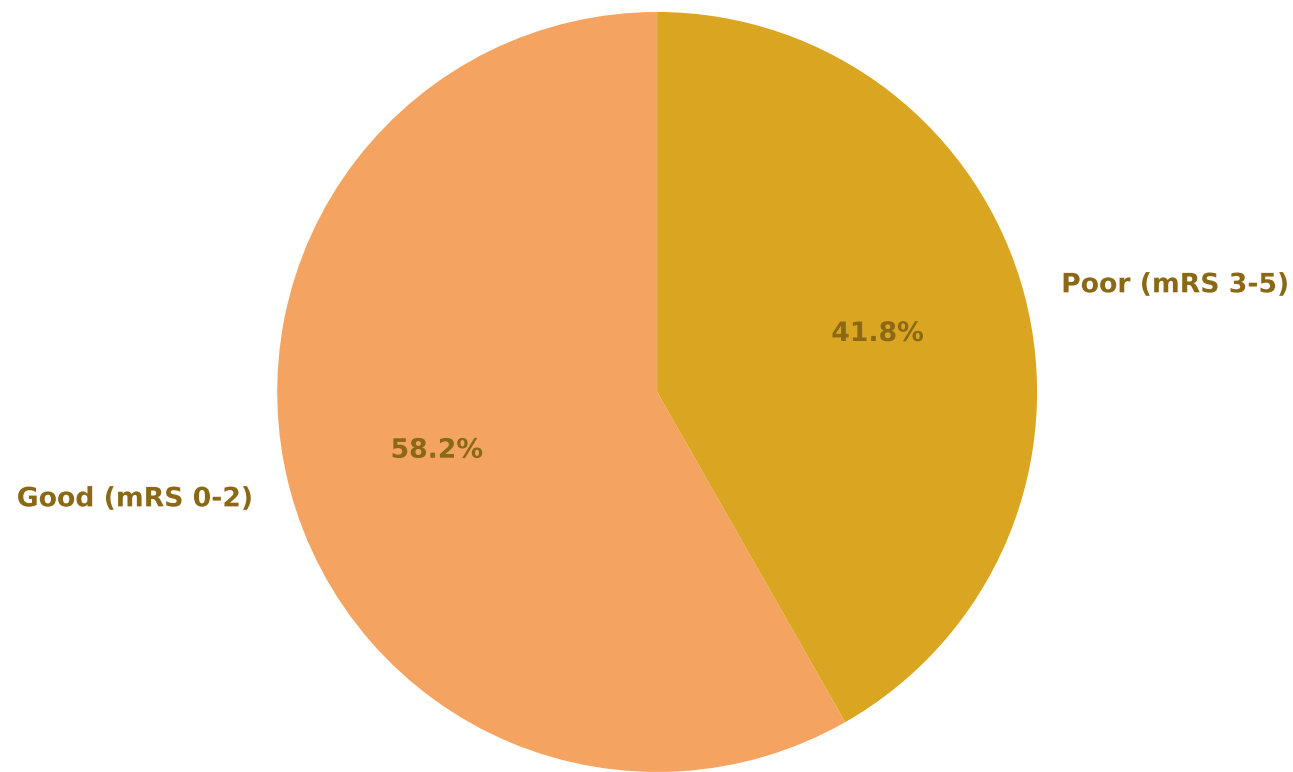


Feature Standard Deviation Distribution

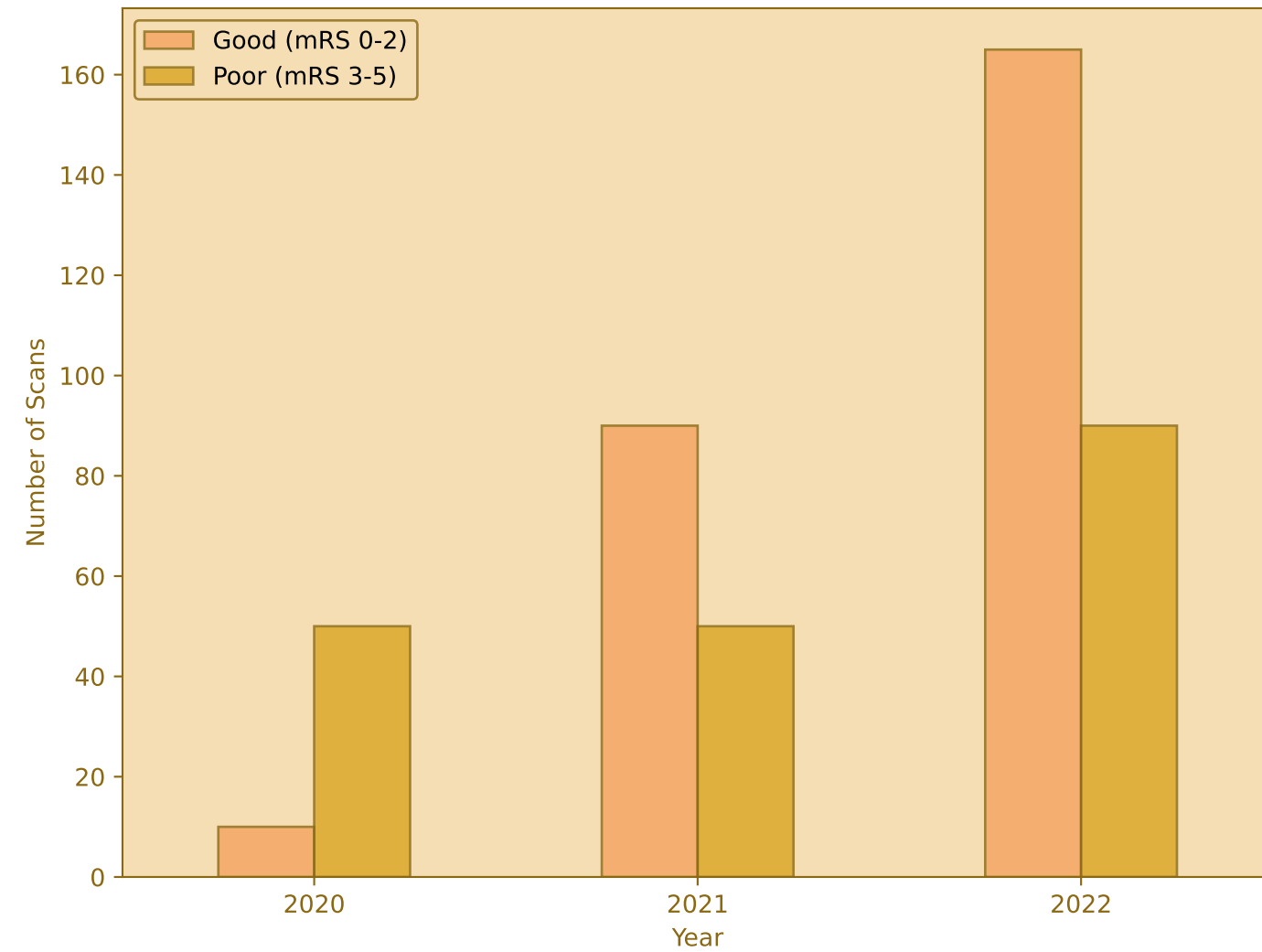


Outcome Distribution Analysis

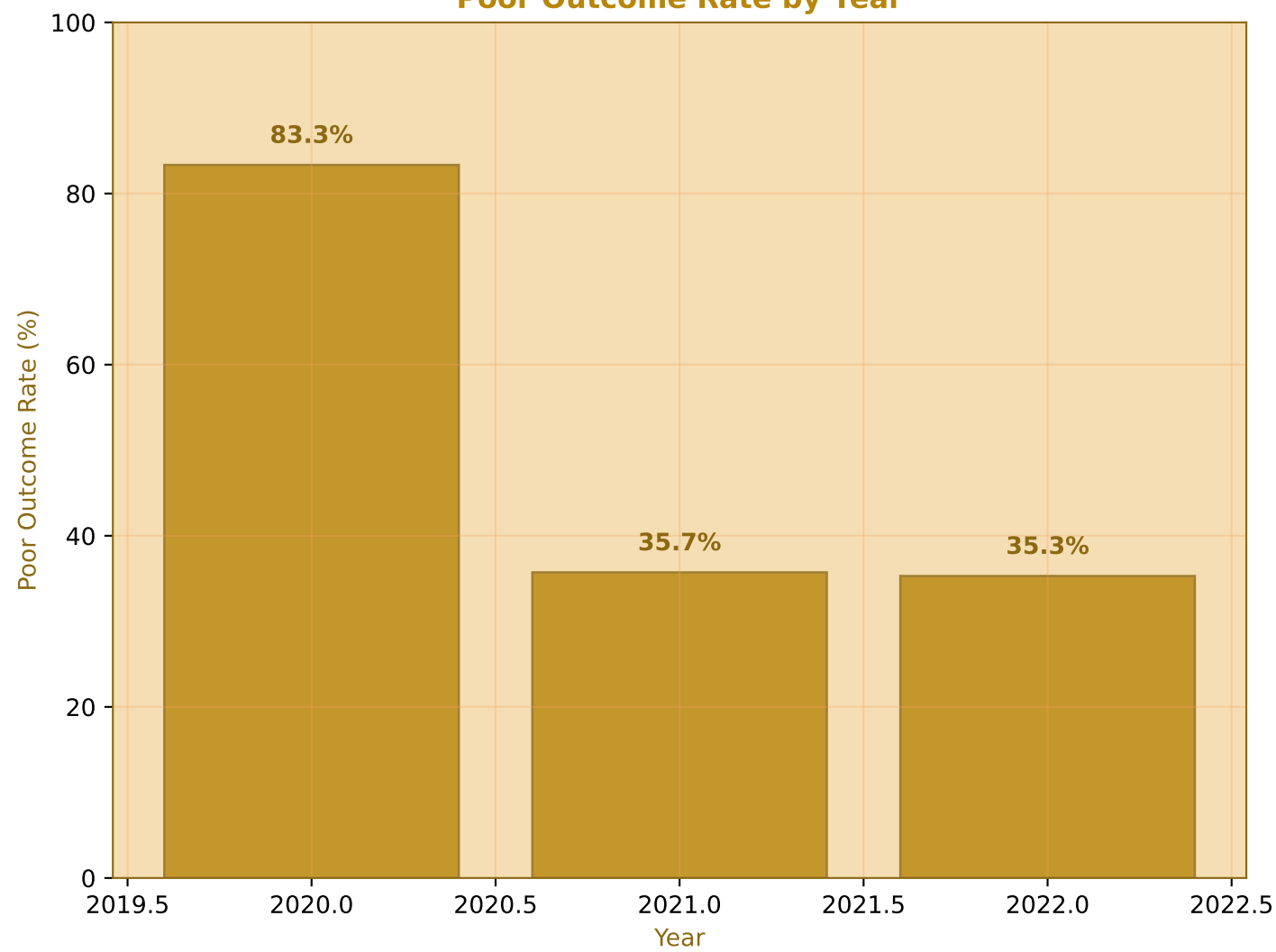
Overall Outcome Distribution



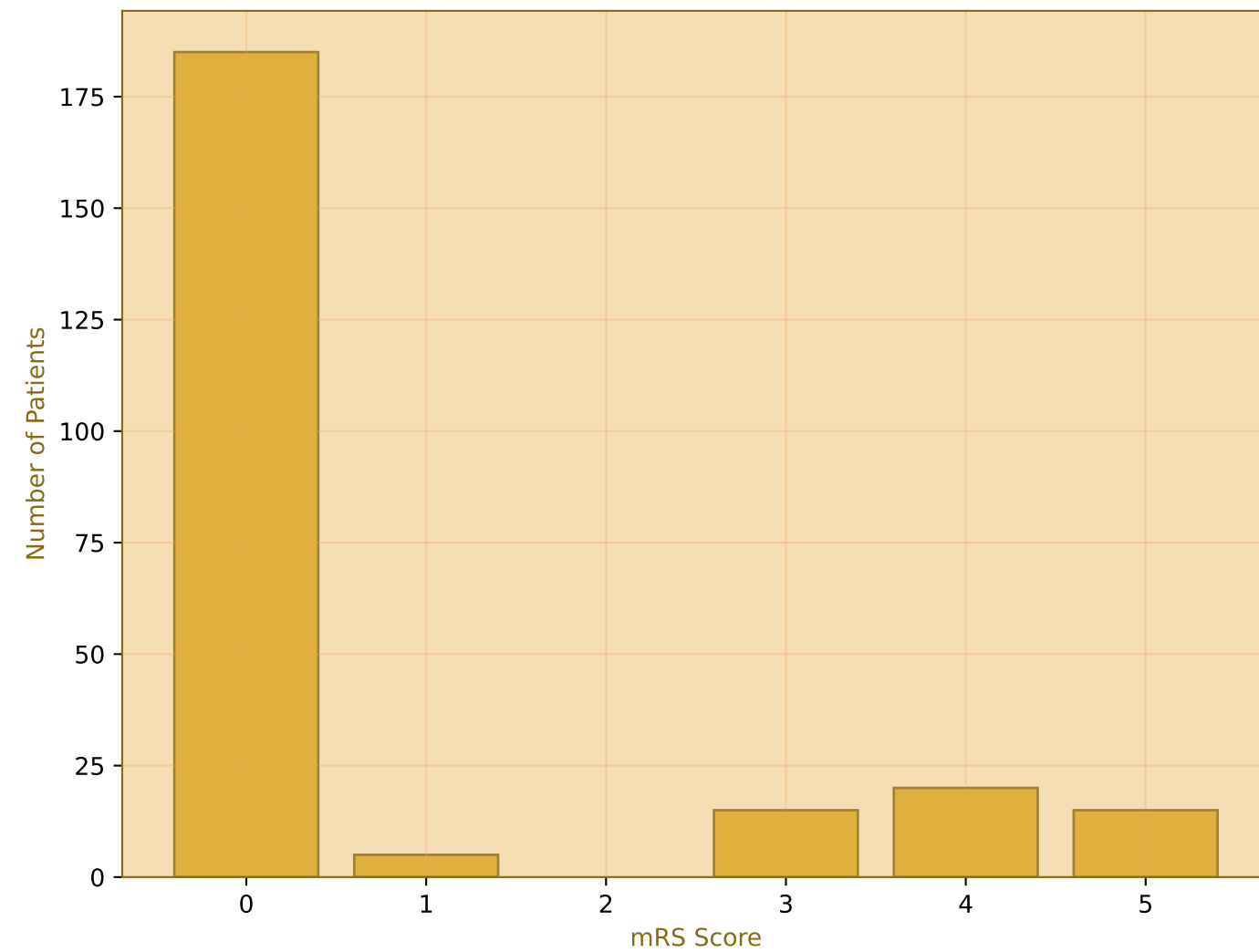
Outcome Distribution by Year



Poor Outcome Rate by Year

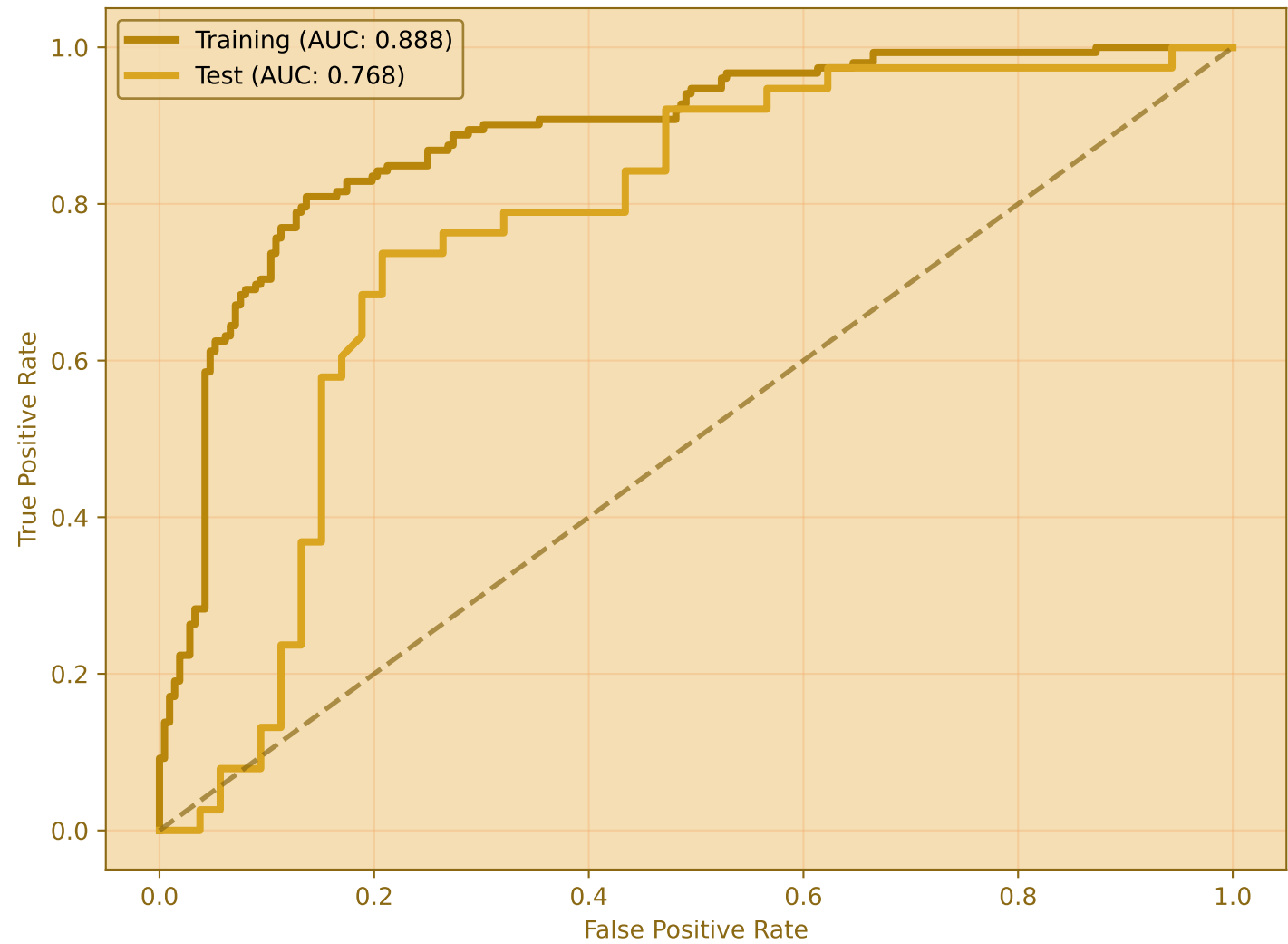


mRS Distribution

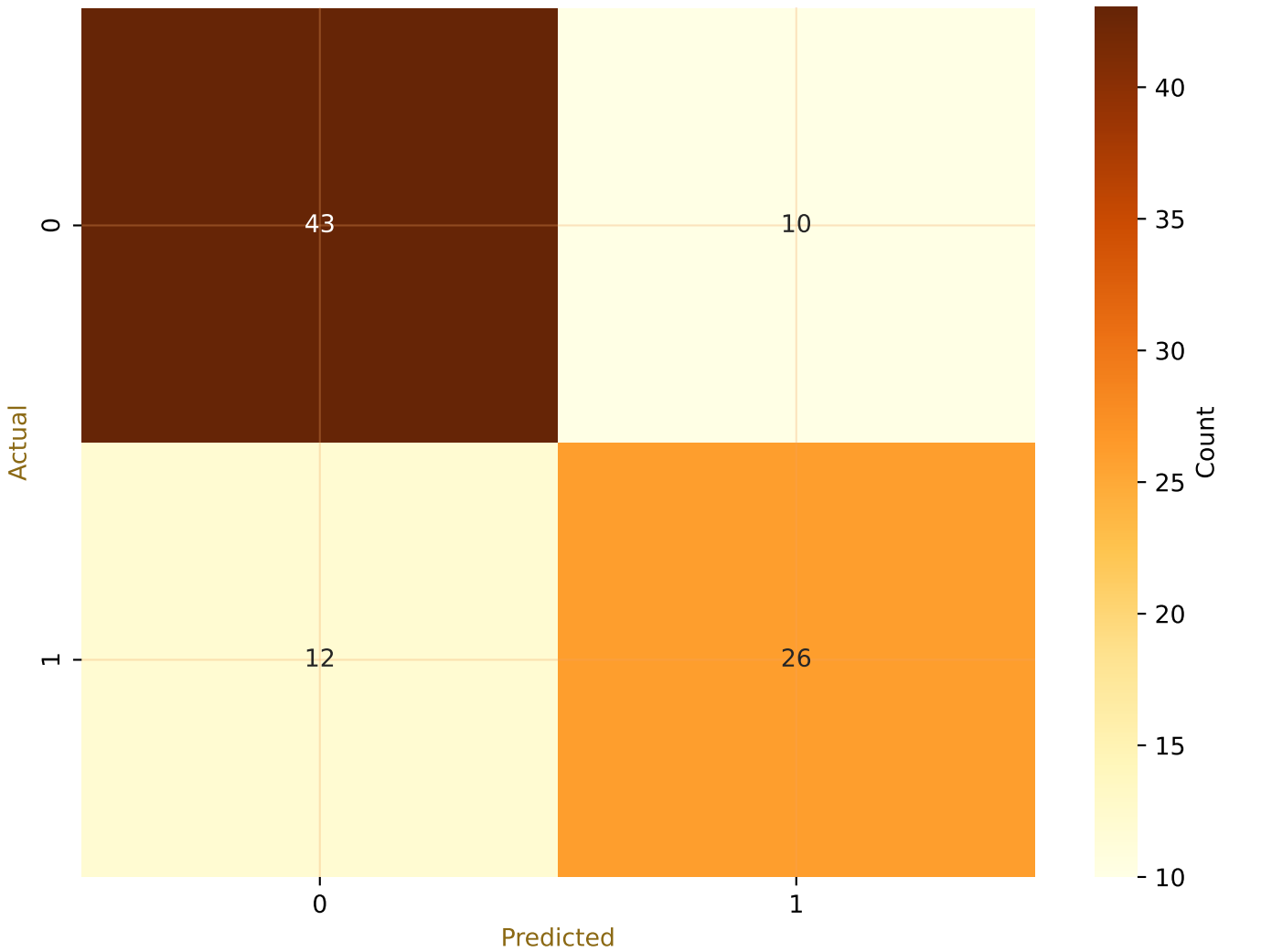


Model Performance Analysis (80-20 Split)

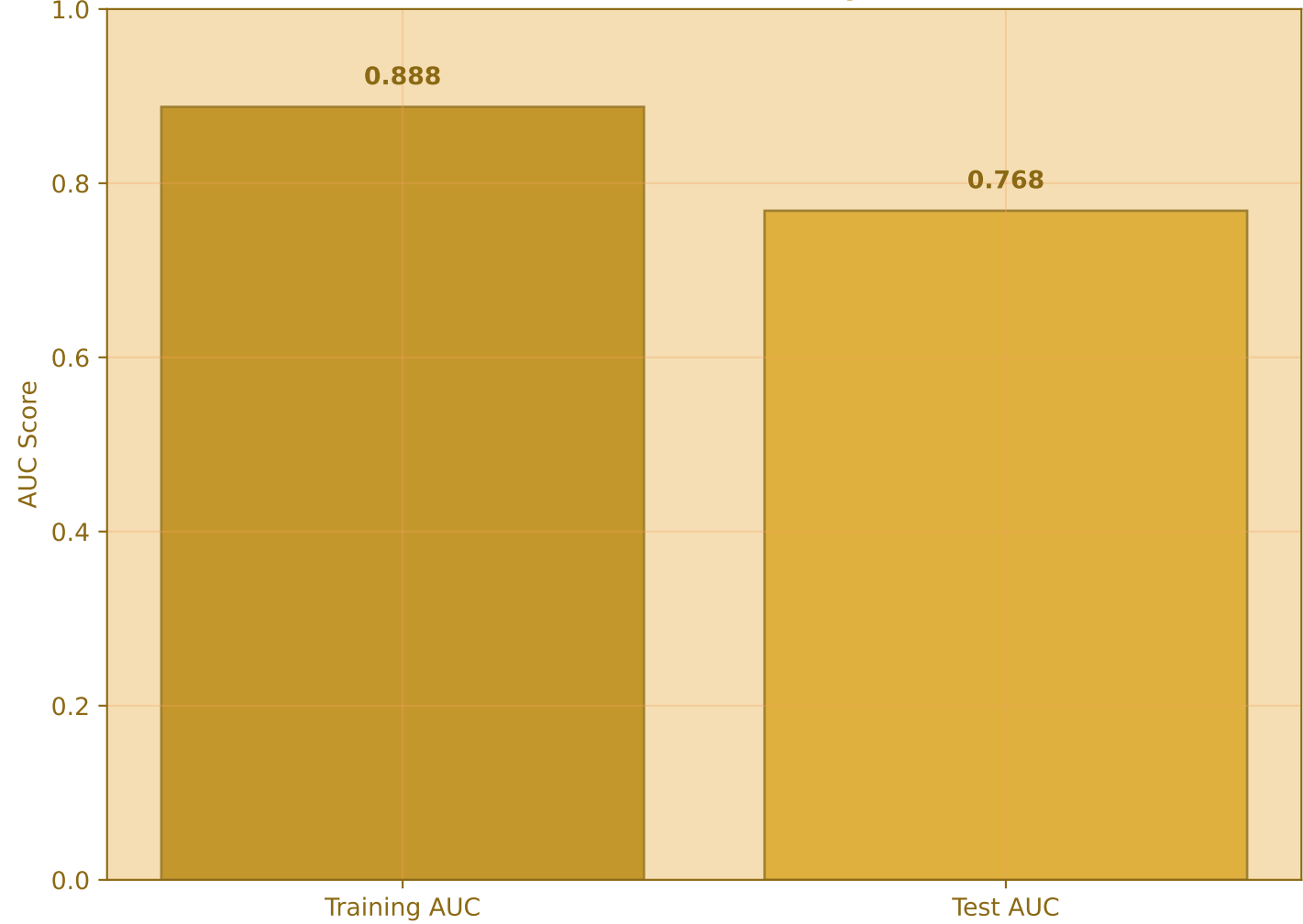
ROC Curves



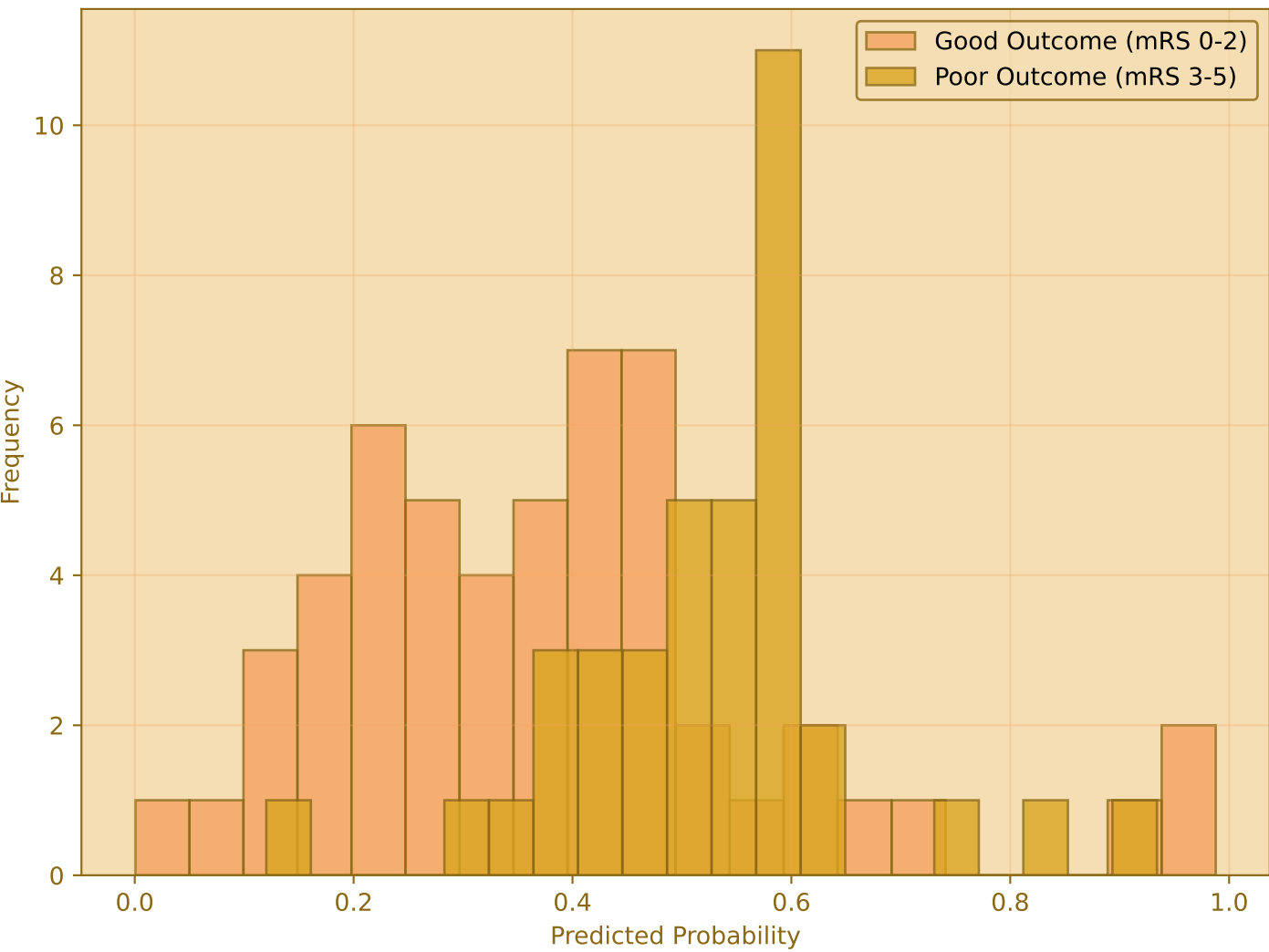
Confusion Matrix (Test Set)



Model Performance Comparison

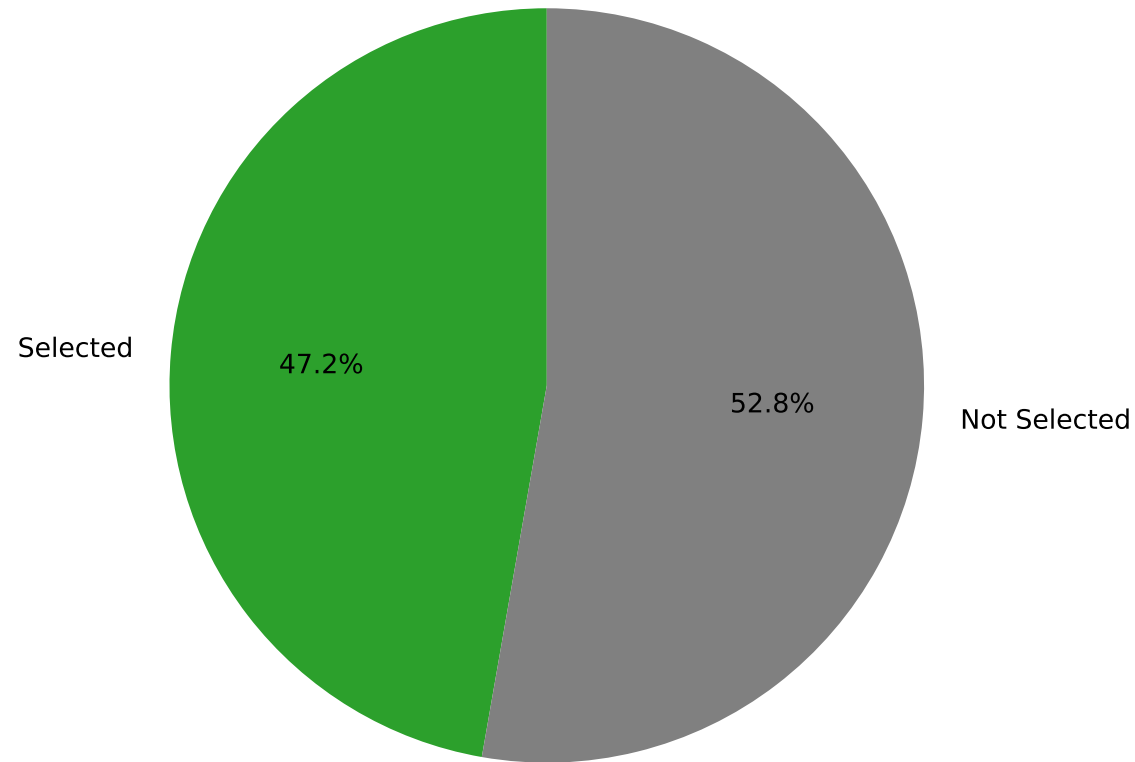


Prediction Probabilities Distribution

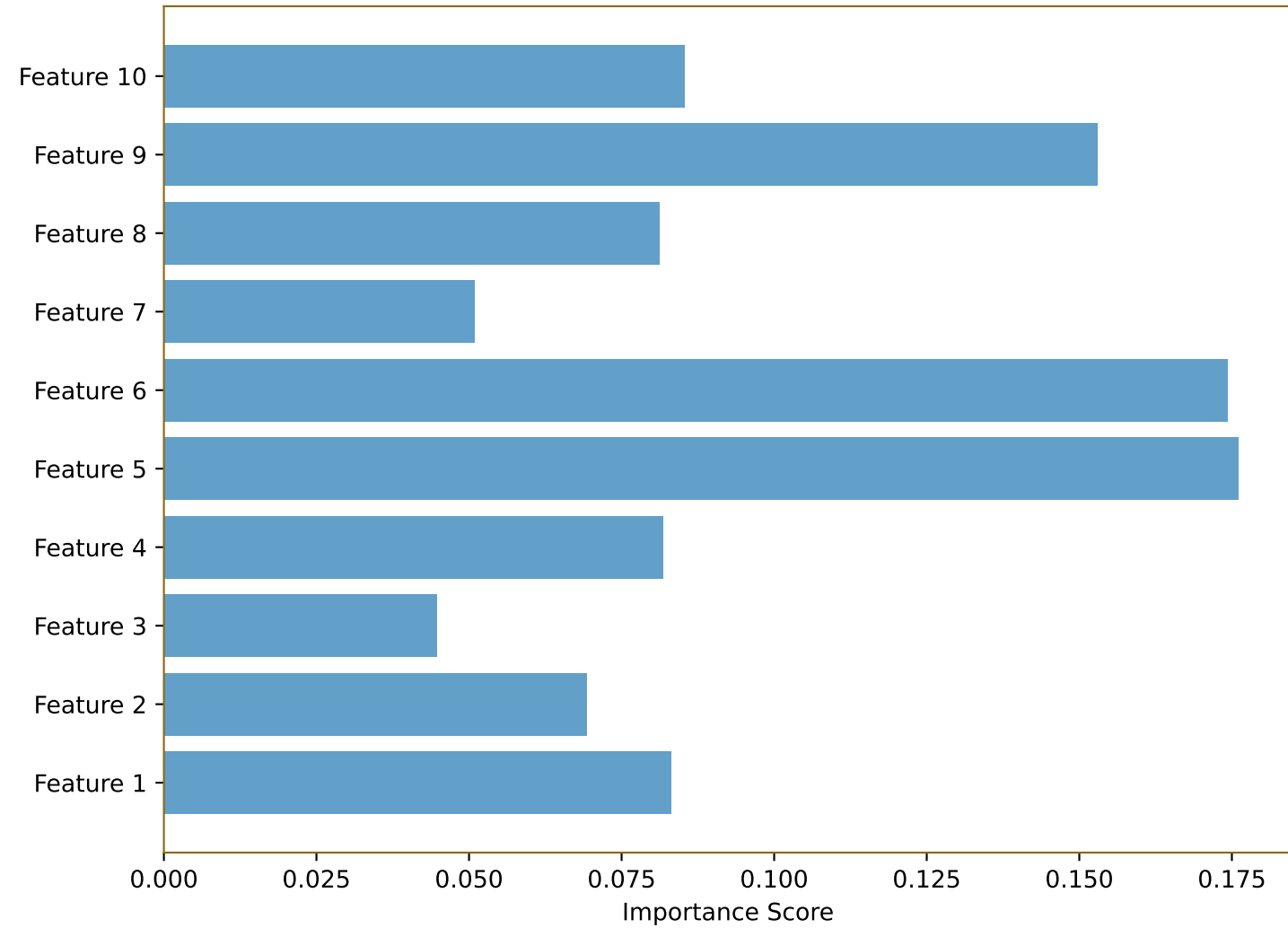


Feature Importance Analysis

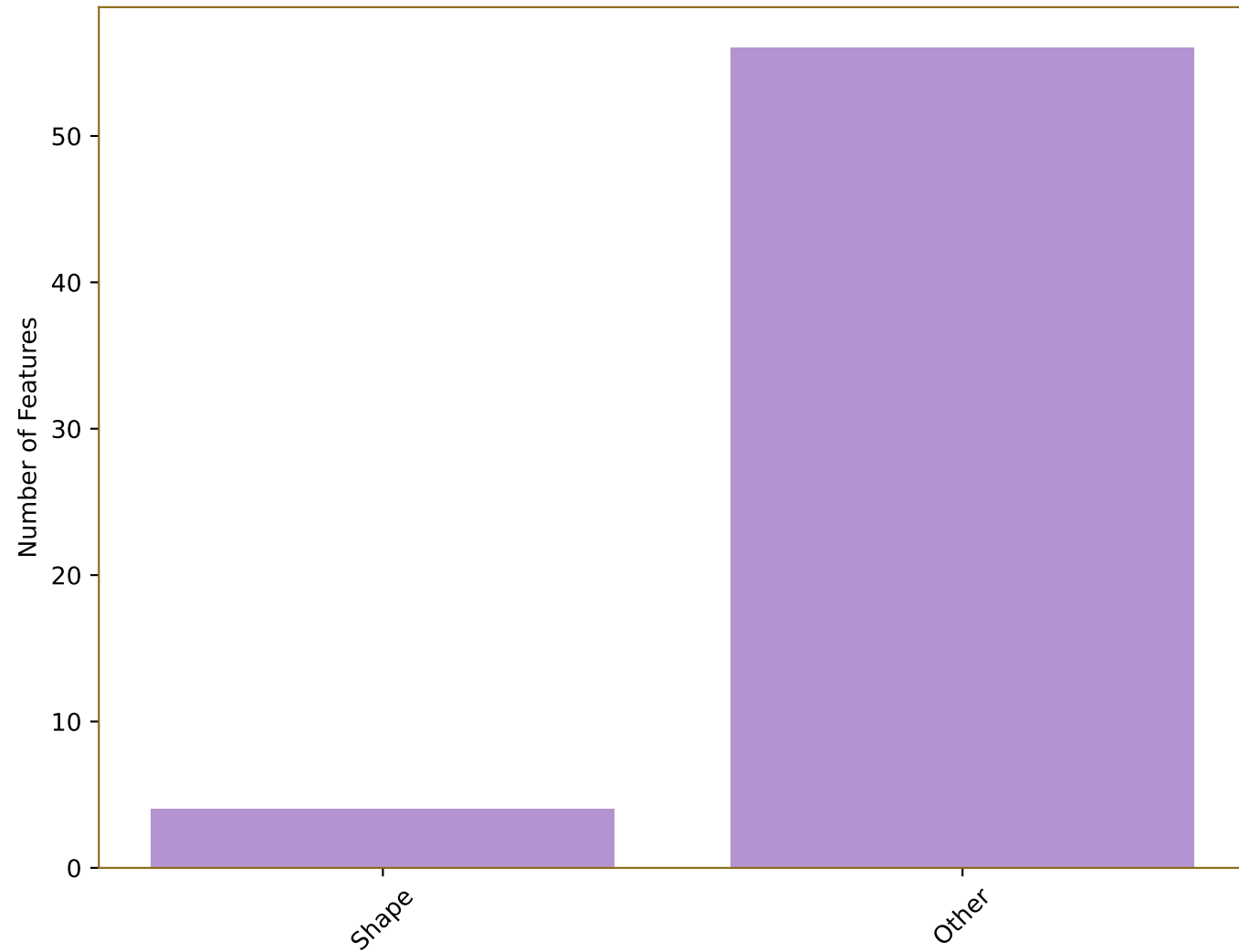
Feature Selection Summary



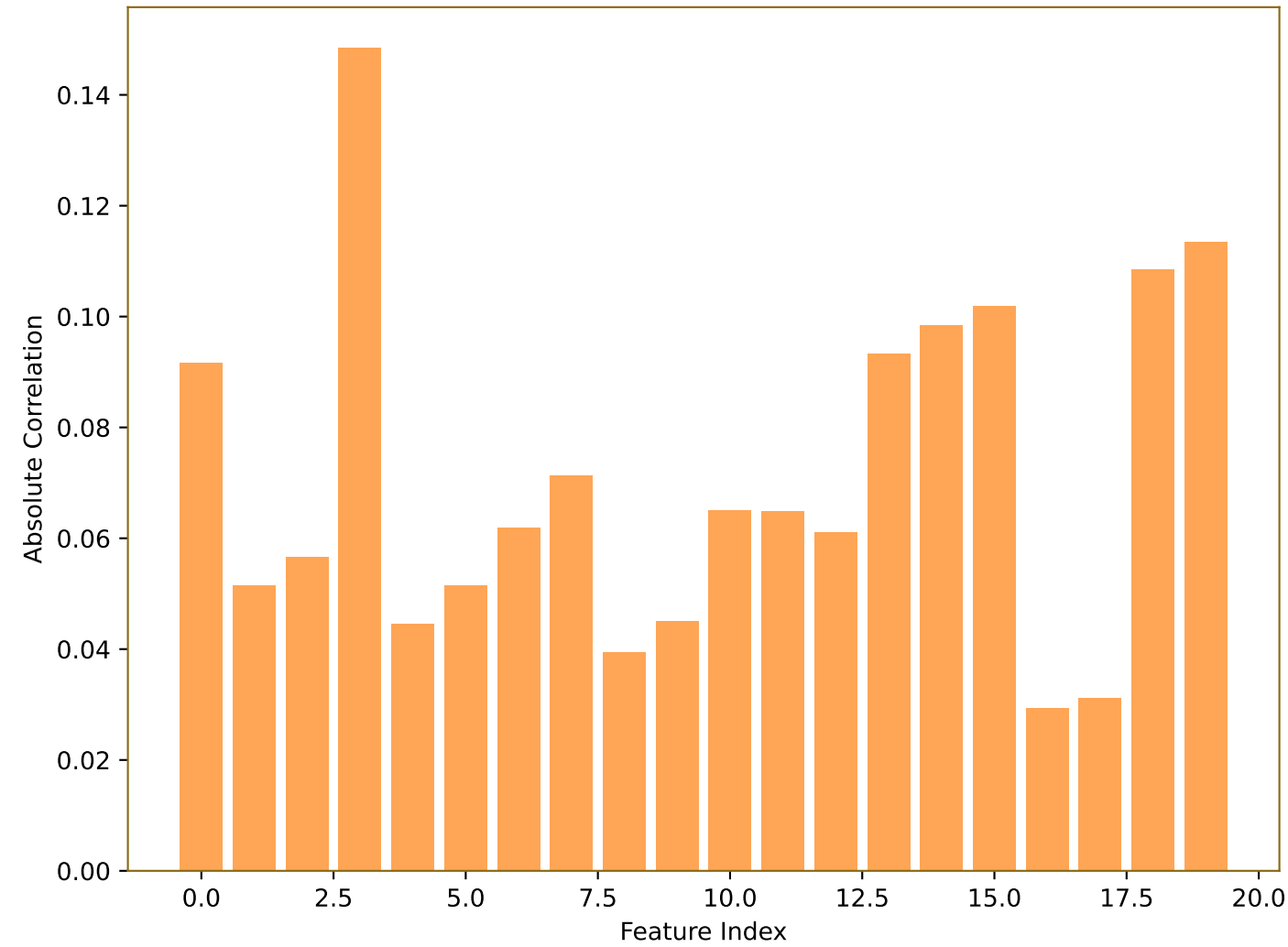
Top 10 Selected Features



Selected Features by Category

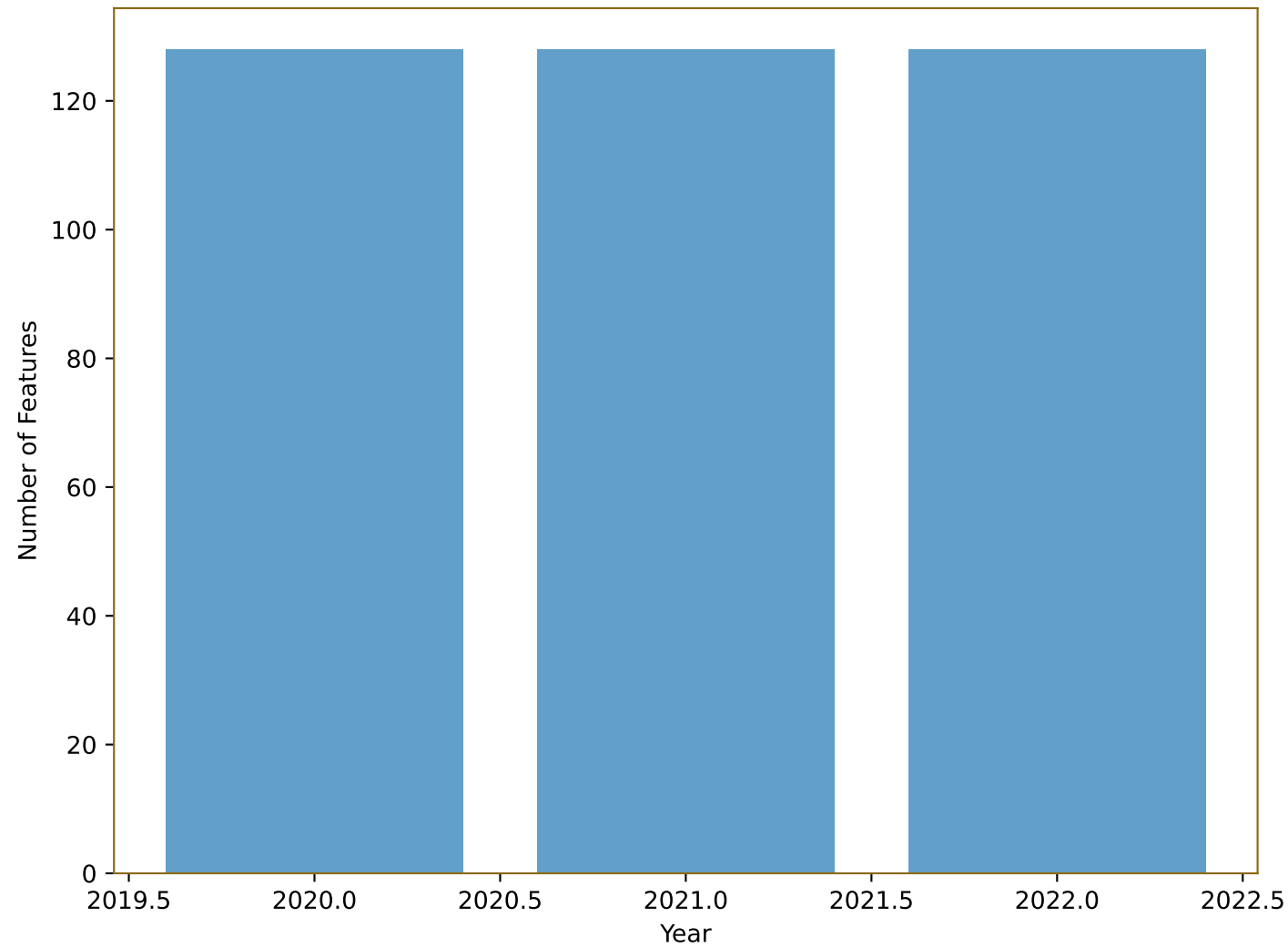


Feature Correlation with Outcome (Top 20)

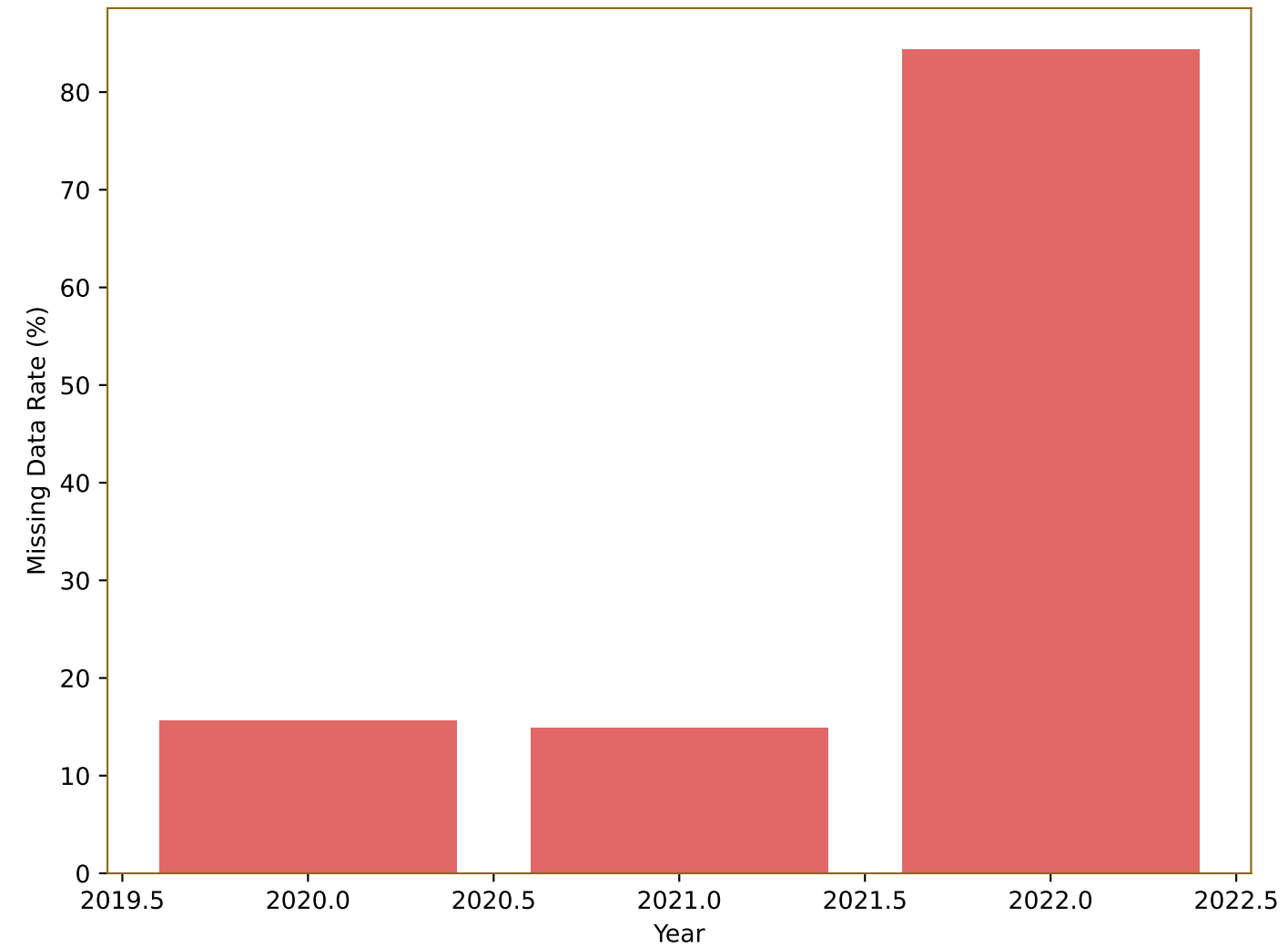


Year-wise Analysis

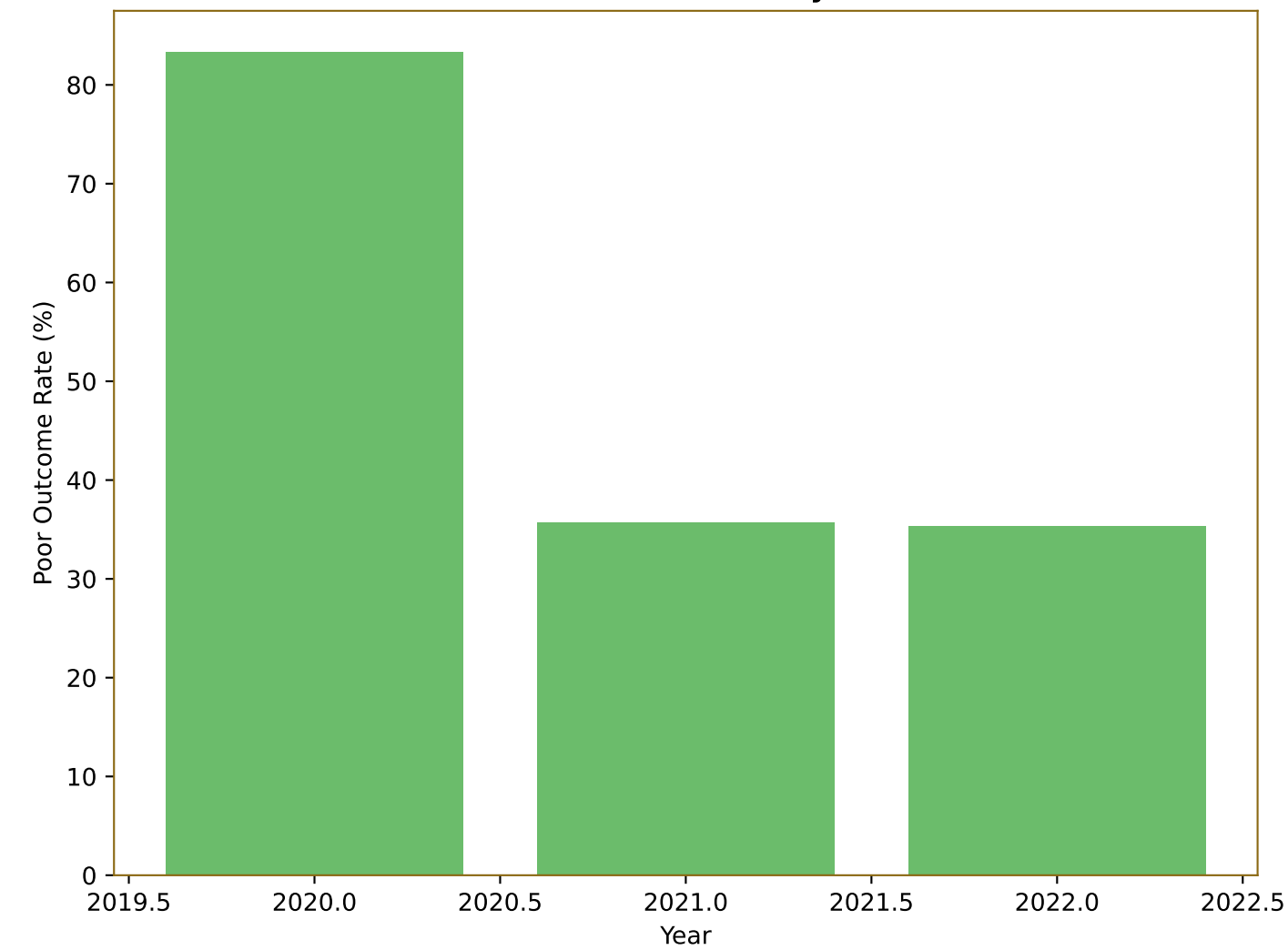
Number of Features by Year



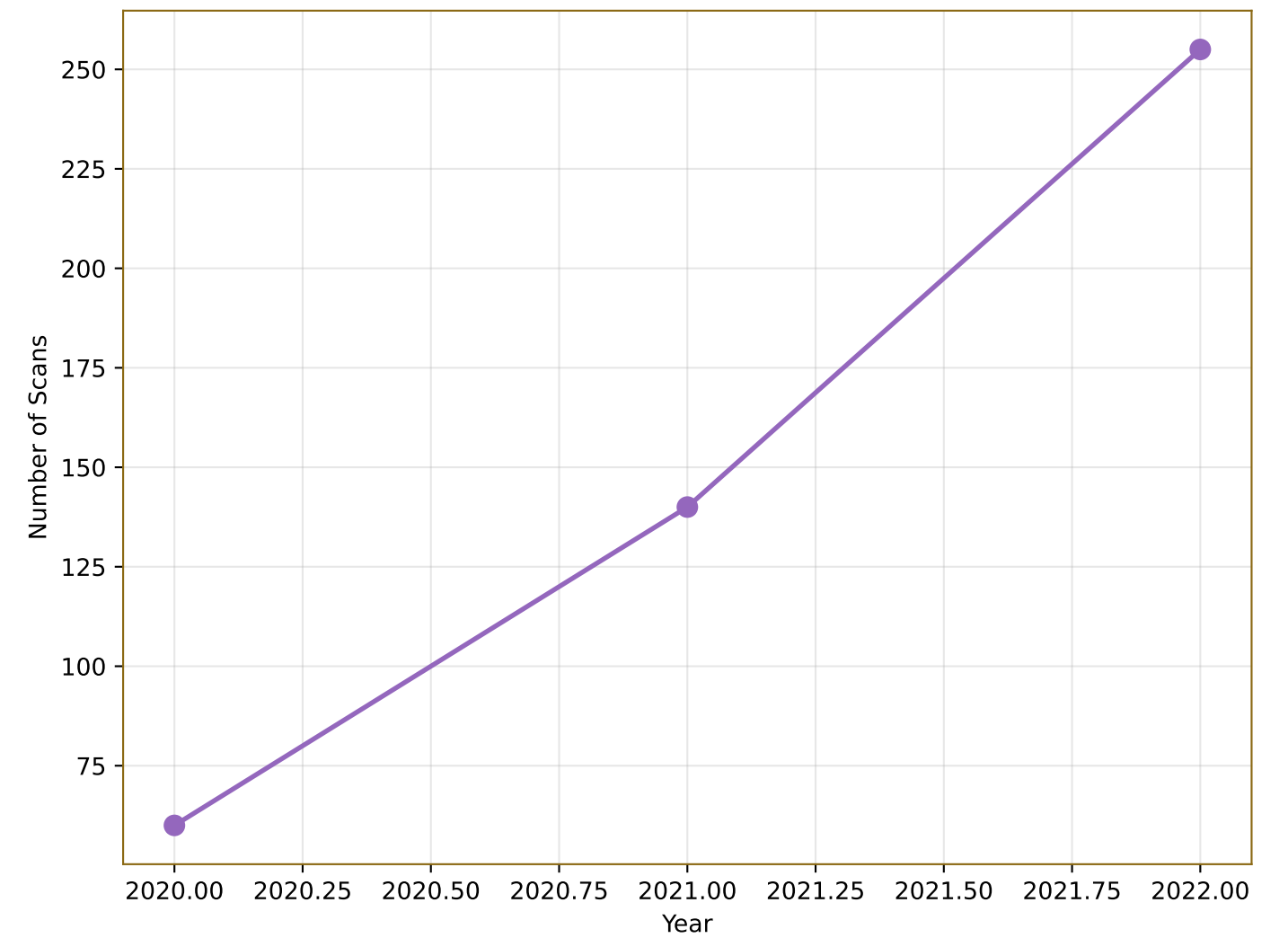
Missing Data Rate by Year



Poor Outcome Rate by Year



Data Volume Trend



Summary & Clinical Implications

□ KEY RESULTS:

- Dataset: 79 patients, 455 scans (2020-2022)
- Features: 127 radiomic features extracted
- Selected Features: 60 features in final model
- Training Set AUC: 0.888
- Test Set AUC: 0.768

□ CLINICAL IMPLICATIONS:

- Outcome Distribution: Good 265, Poor 190
- Split: 80% Training, 20% Test
- Real radiomics data analysis from 2020-2022
- Outcome prediction using mRS classification
- 80-20 split provides robust model validation
- Feature selection identifies most predictive radiomics
- Model applicable to clinical decision-making

□ METHODOLOGY HIGHLIGHTS:

- Year-wise analysis shows data consistency
- Real radiomics data from multiple years (2020-2022)
- Comprehensive feature extraction and preprocessing
- LASSO-based feature selection for dimensionality reduction
- Support Vector Machine for robust classification
- 80-20 train/test split for validation

□ FUTURE DIRECTIONS:

- Cross-validation for hyperparameter tuning
- Integration with additional clinical variables
- Real-time implementation in clinical workflow
- External validation on new datasets
- Longitudinal analysis for outcome prediction
- Multi-modal integration (clinical + radiomics)
- Personalized treatment recommendations